

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Description: Borane-dimethyl sulfide complex, 2M in THF
Cat No. : 42965
Synonyms Dimethyl sulfideborane; BMS
Molecular Formula C2 H9 B S

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

Details of the supplier of the safety data sheet
Importer

Fisher Scientific Korea
 D5,D6, Incheon Airport Logistics Complex
 150, Gonghangdong-Ro 296 Beon-Gil
 Jung-Gu, Incheon
 Tel: +82-1661-9555
 Fax: +82-2-2023-0603

Supplier

Thermo Fisher Scientific Chemicals, Inc.
 30 Bond Street
 Ward Hill, MA 01835-8099

E-mail address Chem.KR@thermofisher.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture
Physical hazards

Flammable liquids	Category 2
Substances/mixtures which, in contact with water, emit flammable gases	Category 1

Health hazards

Acute Oral Toxicity	Category 4
Acute Dermal Toxicity	Category 4
Skin Corrosion/Irritation	Category 2
Serious Eye Damage/Eye Irritation	Category 1
Carcinogenicity	Category 2
Reproductive Toxicity	Category 1
Specific target organ toxicity - (single exposure)	Category 3

Environmental hazards

Based on available data, the classification criteria are not met

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Label Elements



Signal Word

Danger

Hazard Statements

H260 - In contact with water releases flammable gases which may ignite spontaneously
H315 - Causes skin irritation
H225 - Highly flammable liquid and vapor
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H336 - May cause drowsiness or dizziness
H351 - Suspected of causing cancer
H360 - May damage fertility or the unborn child
H302 + H312 - Harmful if swallowed or in contact with skin

Precautionary Statements

Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P233 - Keep container tightly closed
P240 - Ground and bond container and receiving equipment
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment
P242 - Use non-sparking tools
P243 - Take action to prevent static discharges
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P223 - Do not allow contact with water
P231 + P232 - Handle and store contents under inert gas. Protect from moisture
P264 - Wash hands and face thoroughly after handling
P270 - Do not eat, drink or smoke when using this product
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray
P271 - Use only outdoors or in a well-ventilated area
P201 - Obtain special instructions before use
P202 - Do not handle until all safety precautions have been read and understood

Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction
P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water or wrap in wet bandages
P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell
P330 - Rinse mouth
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P312 - Call a POISON CENTER or doctor if you feel unwell
P321 - Specific treatment (see supplemental first aid instructions on this label)
P362 + P364 - Take off contaminated clothing and wash it before reuse
P332 + P313 - If skin irritation occurs: Get medical advice/attention
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P308 + P313 - IF exposed or concerned: Get medical advice/attention

Storage

P403 + P235 - Store in a well-ventilated place. Keep cool

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P402 + P404 - Store in a dry place. Store in a closed container
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
P405 - Store locked up

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

Reacts violently with water
May form explosive peroxides

Stench

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

NFPA

Health
2

Flammability
4

Instability
2

Physical hazards
W

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS No	Index No	Weight %
Tetrahydrofuran	THF; butylene oxide; furanidine; oxolane	109-99-9	KE-33454	80 - 90
Boron, trihydro[thiobis[methane]]-, (T-4)-	BMS; Dimethyl sulfideborane	13292-87-0	2008-1-560	>=10 - 20

SECTION 4: FIRST AID MEASURES

Description of first aid measures

General Advice

If symptoms persist, call a physician.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion

Clean mouth with water and drink afterwards plenty of water.

Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

Most important symptoms and effects, both acute and delayed

Causes severe eye damage. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Causes central nervous system depression.

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically. Symptoms may be delayed.

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SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

CO₂, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons

Water.

Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Reacts violently with water.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Sulfur oxides, Oxides of boron, Hydrogen chloride gas, Hydrogen.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

Environmental precautions

Should not be released into the environment.

Methods and Material for Containment and Cleaning Up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Do not allow contact with water. Handle under an inert atmosphere. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

Conditions for Safe Storage, Including any Incompatibilities

Keep away from heat, sparks and flame. Keep from any possible contact with water. Keep under nitrogen. Keep container tightly closed in a dry and well-ventilated place. Shelf life 12 months. May form explosive peroxides on prolonged storage. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from water or moist air. Store under an inert atmosphere. Protect from

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moisture. Keep refrigerated.

Specific End Uses

Use in laboratories.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
Tetrahydrofuran	109-99-9	STEL: 100 ppm TWA: 50 ppm Skin	TWA: 50 ppm STEL: 100 ppm Skin	(Vacated) TWA: 200 ppm (Vacated) TWA: 590 mg/m ³ (Vacated) STEL: 250 ppm (Vacated) STEL: 735 mg/m ³ TWA: 200 ppm TWA: 590 mg/m ³
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not listed	Not listed	Not listed

Component	CAS No	European Union	The United Kingdom	Germany
Tetrahydrofuran	109-99-9	TWA: 50 ppm (8h) TWA: 150 mg/m ³ (8h) STEL: 100 ppm (15min) STEL: 300 mg/m ³ (15min) Skin	STEL: 100 ppm 15 min STEL: 300 mg/m ³ 15 min TWA: 50 ppm 8 hr TWA: 150 mg/m ³ 8 hr Skin	TWA: 50 ppm (8 Stunden). AGW - exposure factor 2 TWA: 150 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 50 ppm (8 Stunden). MAK TWA: 150 mg/m ³ (8 Stunden). MAK Höhepunkt: 100 ppm Höhepunkt: 300 mg/m ³ Haut
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not listed	Not listed	Not listed

ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
Tetrahydrofuran	109-99-9	2 mg/L Medium: urine Time: end of shift Determinant: Tetrahydrofuran
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not listed

Exposure Controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

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(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Personal protective equipment Use only those certified by the Korea Occupational Safety and Health Administration.
Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators
Recommended Filter type: low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
When RPE is used a face piece Fit Test should be conducted

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Light yellow Liquid

Odor Stench
Odor Threshold No data available
pH No information available

Melting Point/Range No data available
Softening Point No data available
Boiling Point/Range No information available
Flash Point -17 °C / 1.4 °F **Method -** No information available

Evaporation Rate No data available
Flammability (solid,gas) Not applicable Liquid
Explosion Limits No data available

Vapor Pressure No information available
Vapor Density No information available (Air = 1.0)
Specific Gravity / Density 0.850
Bulk Density Not applicable Liquid
Water Solubility Reacts with water
Solubility in other solvents No information available

Partition Coefficient (n-octanol/water)

Component	CAS No	log Pow
Tetrahydrofuran	109-99-9	0.45
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	No data available

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Autoignition Temperature	No data available	
Decomposition Temperature	44 °C	
Viscosity	No data available	
Explosive Properties		Vapors may form explosive mixtures with air
Oxidizing Properties	No information available	
Molecular Formula	C2 H9 B S	
Molecular Weight	75.95	

SECTION 10: STABILITY AND REACTIVITY

Reactivity

Yes Contact with water liberates extremely flammable gases.

Chemical Stability

Reacts violently with water, liberating extremely flammable gases. Stable under normal conditions. Moisture sensitive.

Possibility of Hazardous Reactions

Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	Reacts violently with water.

Conditions to Avoid

Keep away from open flames, hot surfaces and sources of ignition. Incompatible products. Exposure to moist air or water. Exposure to moisture.

Incompatible Materials

Acids. Water. Alcohols. Bromine. Acid anhydrides. Acid chlorides.

Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO₂). Sulfur oxides. Oxides of boron. Hydrogen chloride gas. Hydrogen.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Information on expected route of exposure

Use only in closed systems permitted

Inhalation	Not an expected route of exposure.
Ingestion	May be harmful if swallowed.
Eyes	Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. May cause irritation. Risk of serious damage to eyes.
Skin	Avoid contact with skin. Skin Corrosion/Irritation. May cause irritation. Contact with moist skin may cause skin burns. Harmful in contact with skin.

Information on Health Hazards

(a) acute toxicity;	.
Oral	Category 4

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Dermal
Inhalation

Category 4
Based on available data, the classification criteria are not met

Toxicology data for the components

Component	CAS No	LD50 Oral	LD50 Dermal	LC50 Inhalation
Tetrahydrofuran	109-99-9	1650 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	180 mg/L (Rat) 1 h 53.9 mg/L (Rat) 4 h
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	<500 mg/kg (Rat)	>2000 mg/kg (Rabbit)	No data available

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

Component	CAS No	Test method	Test species	Study result
Tetrahydrofuran	109-99-9	Local Lymph Node Assay OECD Test Guideline 429	mouse	non-sensitising
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	No data available	No data available	No data available

(e) germ cell mutagenicity; No data available

Component	CAS No	Test method	Test species	Study result
Tetrahydrofuran	109-99-9	OECD Test Guideline 476 Gene cell mutation	in vivo Mammalian	negative
		OECD Test Guideline 473 Chromosomal aberration assay	in vitro Mammalian	negative
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	No data available	No data available	No data available

(f) carcinogenicity; Category 2

Component	CAS No	Test method	Test species / Duration	Study result
Tetrahydrofuran	109-99-9	No data available	No data available	No data available
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	No data available	No data available	No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen Limited evidence of a carcinogenic effect

Component	CAS No	IARC	NTP	ACGIH	OSHA	UK
Tetrahydrofuran	109-99-9	Group 2B	Not listed	A3	X	Not listed
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not listed	Not listed	Not listed	Not listed	Not listed

IARC (International Agency for Research on Cancer)

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

A1 - Known Human Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

(g) reproductive toxicity; Category 1

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Component	CAS No	Test method	Test species / Duration	Study result
Tetrahydrofuran	109-99-9	OECD Test Guideline 416	Rat 2 Generation	NOAEL = 3,000 ppm
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	No data available	No data available	No data available

(h) STOT-single exposure;
Results / Target organs

Category 3
Respiratory system, Central nervous system (CNS).

(i) STOT-repeated exposure;
Target Organs

No data available
None known.

(j) aspiration hazard;

No data available

Other Adverse Effects

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Causes central nervous system depression.

Endocrine Disruptor Information

Component	CAS No	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
Tetrahydrofuran	109-99-9	Group III Chemical	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

Component	CAS No	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Tetrahydrofuran	109-99-9	2160 mg/l LC50 = 96 h Pimephales promelas Leuciscus idus: LC50: 2820 mg/L/48h	EC50 48 h 3485 mg/l EC50: >10000 mg/L/24h	No data available	No data available
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	No data available	No data available	No data available	No data available

Persistence and degradability

Persistence
Degradability
Degradation in sewage treatment plant

Persistence is unlikely.
Reacts with water.
Water reactive.

Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Tetrahydrofuran	0.45	No data available

Mobility in soil

The product is water soluble, and may spread in water systems Reacts with water . Will likely be mobile in the environment due to its water solubility. Is not likely mobile in the environment. Highly mobile in soils.

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Ozone Depletion Potential

Component	CAS No	Ozone Depletion Potential
Tetrahydrofuran	109-99-9	Not listed
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not listed

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

UN-No	UN3399
Proper Shipping Name	ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE
Technical Shipping Name	Tetrahydrofuran, Boron, trihydro[thiobis[methane]]-, (T-4)-
Hazard Class	4.3
Subsidiary Hazard Class	3
Packing Group	I

IATA

UN-No	UN3399
Proper Shipping Name	Organometallic substance, liquid, water-reactive, flammable
Technical Shipping Name	Tetrahydrofuran, Boron, trihydro[thiobis[methane]]-, (T-4)-
Hazard Class	4.3
Subsidiary Hazard Class	3
Packing Group	I

IMDG/IMO

UN-No	UN3399
Proper Shipping Name	ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE
Technical Shipping Name	Tetrahydrofuran, Boron, trihydro[thiobis[methane]]-, (T-4)-
Hazard Class	4.3
Subsidiary Hazard Class	3
Packing Group	I
Marine Pollutant	No hazards identified

Special Precautions for User

No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

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International Inventories

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	ISHL	AICS
Tetrahydrofuran	109-99-9	KE-33454	X	203-726-8	X	X	-	X	X	X	X
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	2008-1-560	X	236-313-6	X	-	X	-	-	X	-

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Tetrahydrofuran	109-99-9	Not applicable	Not applicable	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Tetrahydrofuran	109-99-9	Listed	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Tetrahydrofuran	109-99-9	Annex 1 - KE-33454	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Annex 2 - 2008-1-560	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Tetrahydrofuran	109-99-9	Not applicable	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	2008-1-0560 (>25%)	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Tetrahydrofuran	109-99-9	Not applicable	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Tetrahydrofuran	109-99-9	Not applicable	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	> 25% (CCA)	Not applicable	Not applicable

CCA = Chemical Control Act

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Tetrahydrofuran	109-99-9	Listed	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable

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Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Tetrahydrofuran	109-99-9	Listed	Listed	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Tetrahydrofuran	109-99-9	5000 kg	STEL: 100 ppm TWA: 50 ppm Skin	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	5000 kg	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Tetrahydrofuran	109-99-9	Not applicable	Not applicable	Not applicable	2. Group 1 Petroleum (Soluble) 400 L	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable	2. Group 1 Petroleum (Insoluble) 200 L	Not applicable	Not applicable

Control Parameters

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
Tetrahydrofuran	109-99-9	STEL: 100 ppm TWA: 50 ppm Skin	2 mg/L Medium: urine Time: end of shift Determinant: Tetrahydrofuran
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not listed	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Tetrahydrofuran	109-99-9	Not applicable	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355)

Component	CAS No	CERCLA Extremely Hazardous Substances RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Tetrahydrofuran	109-99-9	Not applicable	1000 lb	Not applicable
Boron, trihydro[thiobis[methane]]-, (T-4)-	13292-87-0	Not applicable	Not applicable	Not applicable

GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

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Danger.

H225 - Highly flammable liquid and vapor. H260 - In contact with water releases flammable gases which may ignite spontaneously. H315 - Causes skin irritation. H318 - Causes serious eye damage. H335 - May cause respiratory irritation. H336 - May cause drowsiness or dizziness. H351 - Suspected of causing cancer. H360FD - May damage fertility. May damage the unborn child. H302 + H312 - Harmful if swallowed or in contact with skin. EUH014 - Reacts violently with water. EUH019 - May form explosive peroxides.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. P231 + P232 - Handle and store contents under inert gas. Protect from moisture. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 - Immediately call a POISON CENTER or doctor/physician. P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing.

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

Prepared By

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6

Revision Summary

New emergency telephone response service provider.

SAFETY DATA SHEET

Borane-dimethyl sulfide complex, 2M in THF

Revision Date 14-Jun-2024

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

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End of Safety Data Sheet